Asymmetric Effects of Policy Deviations: a Case of RMB Exchange Rate

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RMB Exchange Rate Policies

- PBoC announces central parity rate (CPR) of CNY/USD 9:15 am each business day.
- Intraday spot rate fluctuates within a narrow band around CPR.

▸ CPR and spot rates

- Since August 2015, the PBoC has implemented several reforms to make the formation mechanism of CPR more transparent and market-oriented.

RMB Exchange Rate Policies

- Clark (2017), Su and Qian (2020), Jermann, Wei and Yue (2022)
- CPR is mainly determined by two factors:
  - Market pillar: prior-day CNY/USD close rate
  - Basket pillar: the RMB index (against a basket of currencies)
- CPR does deviate from the two-pillar policy rule
  - random errors or policy interventions?
- This paper: CPR deviations
  - **Predict future exchange rate changes**
  - **Affect surveyed forecast revisions**
  - Both effects **asymmetric** in sign of CPR deviation

Why Market Participants' Surveyed Forecast?

Real Effect and Predictive Power

- Stavrakeva and Tang (2020): USD against 7 currencies
  - Asset position of foreign exchange rate market participant consistent with survey-based forecasts, NOT with FIRE expectations
  - Survey-based forecasts and foreign exchange rate derivative positions explain 70%-80% of quarterly and annually exchange rate changes
- Candian and De Leo (2023): USD against 8 currencies
  - Surveyed forecast errors responsible for UIP puzzle and its reversal at longer horizons
- This paper: CNY/USD

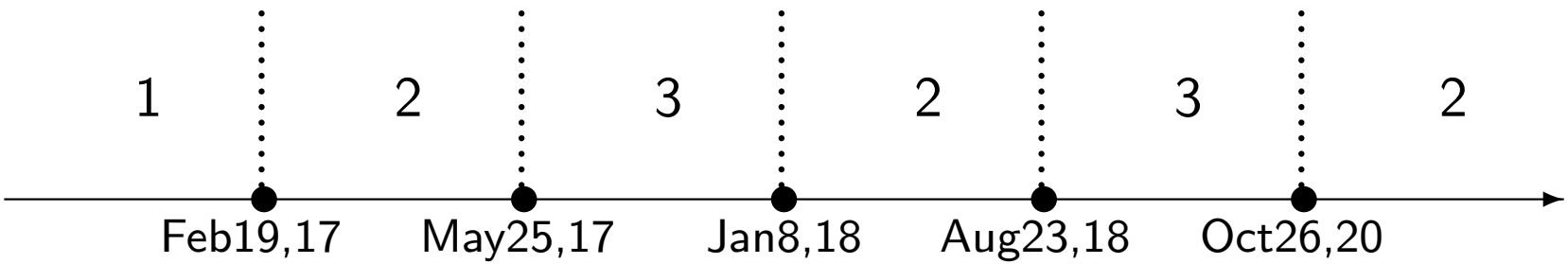
  Surveyed forecast revision predicts future exchange rate changes

Sample: CPR deviation

- 01/25/2016 - 05/28/2021 daily
- Forecast error based on estimated policy rule

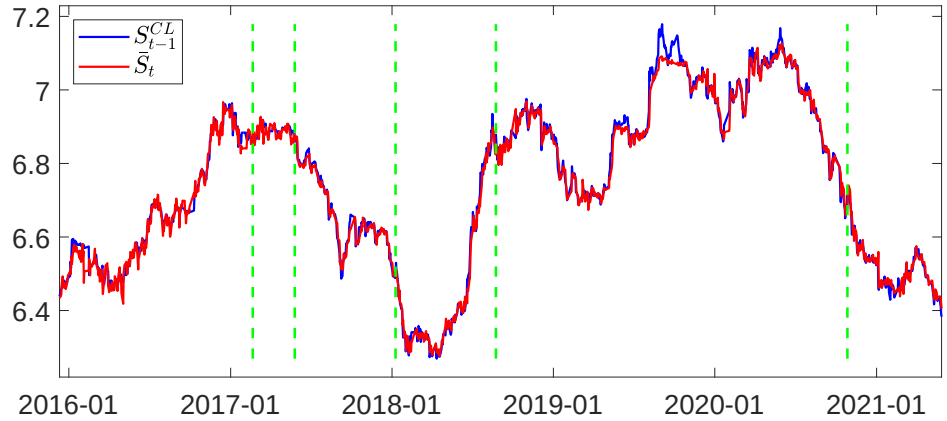
$$S_{t+1}^{CPR} = (\underbrace{S_t^{CL}}_{\text{market pillar}})^{\alpha}(\underbrace{\overline{S}_{t+1}}_{\text{basket pillar}})^{\beta} \quad (1)$$

- 30 days rolling window
- Regime 1: 24-hours basket
- Regime 2: 15-hours nighttime basket
- Regime 3: 15-hours nighttime basket and 9-hours daytime basket

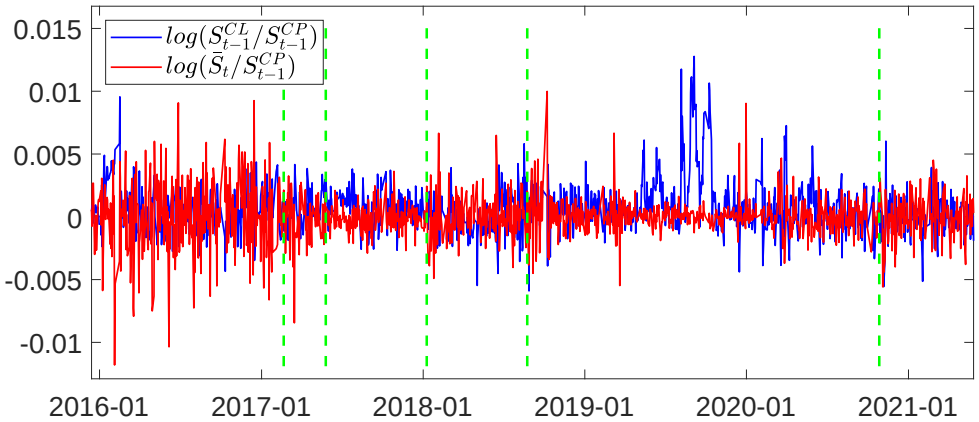


Sample: CPR deviation

Panel A. RMB market closing rates and the basket pillar



Panel B. The *scaled* market closing rates and the basket pillar



Note: S_t^{CP} and S_t^{CL} are the parity rate and the market closing rate, $\bar{S}_t$ is the basket pillar.

Series of CPR deviation

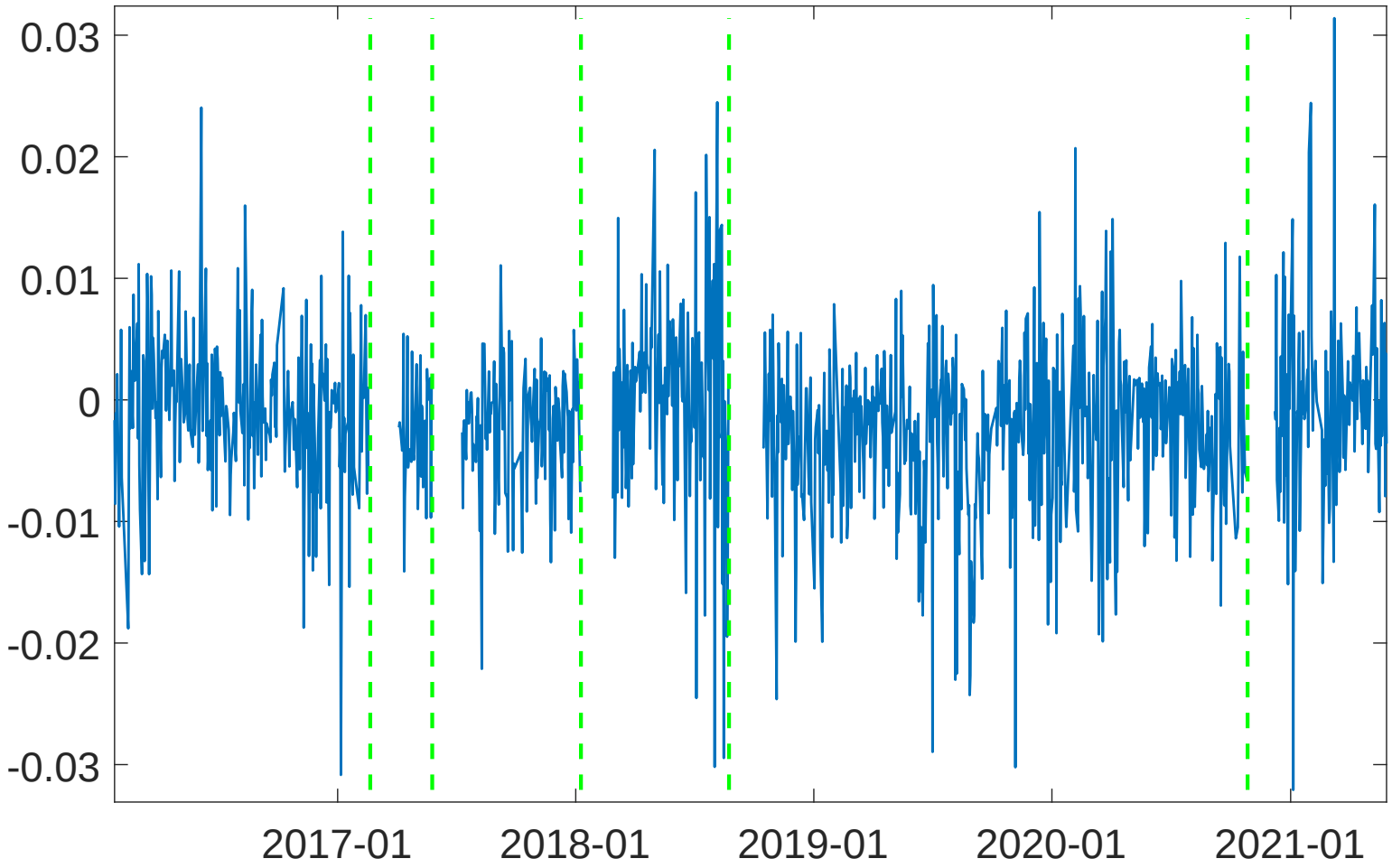


Figure: 30-day rolling window CPR deviation

relative to Close dev

CPR deviation predicts future close rate?

Yes! Sign of CPR deviation matters!

$$S_{t+j}^{CL} - S_{t-1}^{CL} = \alpha + \beta \times \widetilde{CPR}_t + \mathbb{Z}_t\gamma' + \varepsilon_t; \tag{2}$$

- $S_{t+j}^{CL} - S_{t-1}^{CL}$: close rate change j-day ahead, $j = 0, ..., 119$;
- $\widetilde{CPR}_t$: CPR deviation;
- $\mathbb{Z}_t$: S_{t-1}^{CL}, changes in pillars at t

CPR deviation predicts future close rate

dln_close_Xd	1d	2d	3d	4d	5d	10d
			Panel A: all sample			
deviation	0.040***	0.055***	0.061***	0.049**	0.048*	0.077*
	(2.60)	(2.92)	(2.84)	(1.99)	(1.79)	(1.94)
Observations	1136	1135	1134	1133	1132	1127
R-square	0.031	0.030	0.021	0.017	0.019	0.027
			Panel B: deviation<0			
deviation	0.030	-0.002	0.002	-0.022	-0.030	-0.060
	(1.13)	(-0.08)	(0.05)	(-0.50)	(-0.61)	(-0.90)
Observations	682	682	681	681	681	677
R-square	0.019	0.015	0.009	0.009	0.008	0.016
			Panel C: deviation>0			
deviation	0.033	0.101**	0.118**	0.118**	0.148**	0.202**
	(0.91)	(2.26)	(2.15)	(2.05)	(2.38)	(2.29)
Observations	454	453	453	452	451	450
R-square	0.028	0.036	0.023	0.025	0.038	0.039

▸ ruleoutother

CPR deviation predicts future close rate

dln_close_Xd	20d	30d	40d	50d	60d	120d
			Panel A: all sample			
deviation	0.097	0.207***	0.226**	0.209**	0.202*	-0.086
	(1.62)	(2.59)	(2.37)	(2.01)	(1.80)	(-0.59)
Observations	1117	1107	1097	1087	1077	1019
R-square	0.057	0.099	0.145	0.191	0.236	0.461
			Panel B: deviation<0			
deviation	-0.120	-0.105	-0.110	-0.130	-0.218	-0.049
	(-1.27)	(-0.83)	(-0.73)	(-0.80)	(-1.25)	(-0.21)
Observations	673	669	664	659	653	621
R-square	0.030	0.064	0.095	0.119	0.144	0.345
			Panel C: deviation>0			
deviation	0.344**	0.531**	0.684**	0.618**	0.925***	0.487
	(2.32)	(2.52)	(2.49)	(2.02)	(3.23)	(1.37)
Observations	444	438	433	428	424	398
R-square	0.100	0.134	0.206	0.283	0.363	0.601

▸ ruleoutother

Surveyed Forecasts

- Bloomberg open survey on CNY/USD forecast
- 114 professional forecasters from large financial institutions
- quarterly forecast up to next 4-quarter ahead
- annual forecast up to next 5-year ahead
- 31 forecast targets: 2016/1Q to 2025/4Q
- can revise forecast up to one month before the target
- our data covers the entire revision history (daily)
- Average revision frequency: 32 days

Forecast revision predicts future close rate?

$$S^{CL}_{t+j} - S^{CL}_{t-1} = \alpha + \beta \sum_{T,i} \frac{F^i_{T|t} - F^i_{T|t-1}}{N_T N_i} + \mathbb{Z}_t \gamma' + \varepsilon_t; \tag{3}$$

- $S^{CL}_{t+j} - S^{CL}_{t-1}$: close rate change j-day ahead, $j = 0, \ldots, 120$;
- $\sum_{T,i} \frac{F^i_{T|t} - F^i_{T|t-1}}{N_T N_i}$: average FR across forecasters and targets;
- $\mathbb{Z}_t$: S^{CL}_{t-1}, average forecast horizon (T-t)

Forecast revision predicts future close rate

dln_close_Xd	1d	2d	3d	4d	5d	10d
Forecast revision	**0.704*** **	**0.736*** **	**0.731*** **	**0.740*** **	**0.837*** **	**1.135*** **
	(6.31)	(7.13)	(8.27)	(10.17)	(10.33)	(7.47)
length	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1148	1147	1146	1145	1144	1139
R-square	0.135	0.079	0.058	0.049	0.052	0.058

dln_close_Xd	20d	30d	40d	50d	60d	120d
Forecast revision	**1.813*** **	**2.171*** **	**2.350*** **	**2.533*** **	**2.474*** **	**1.879**
	(6.12)	(4.91)	(4.73)	(4.46)	(3.92)	(1.47)
length	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1129	1119	1109	1099	1089	1029
R-square	0.091	0.129	0.177	0.233	0.289	0.465

Take Stock

- $\widetilde{CPR}$ predicts future close rate, with asymmetric effect
  - $\widetilde{CPR} > 0$ predicts higher future close rate (RMB depreciates)
  - $\widetilde{CPR} < 0$ little prediction power
- Surveyed FR predicts future close rate

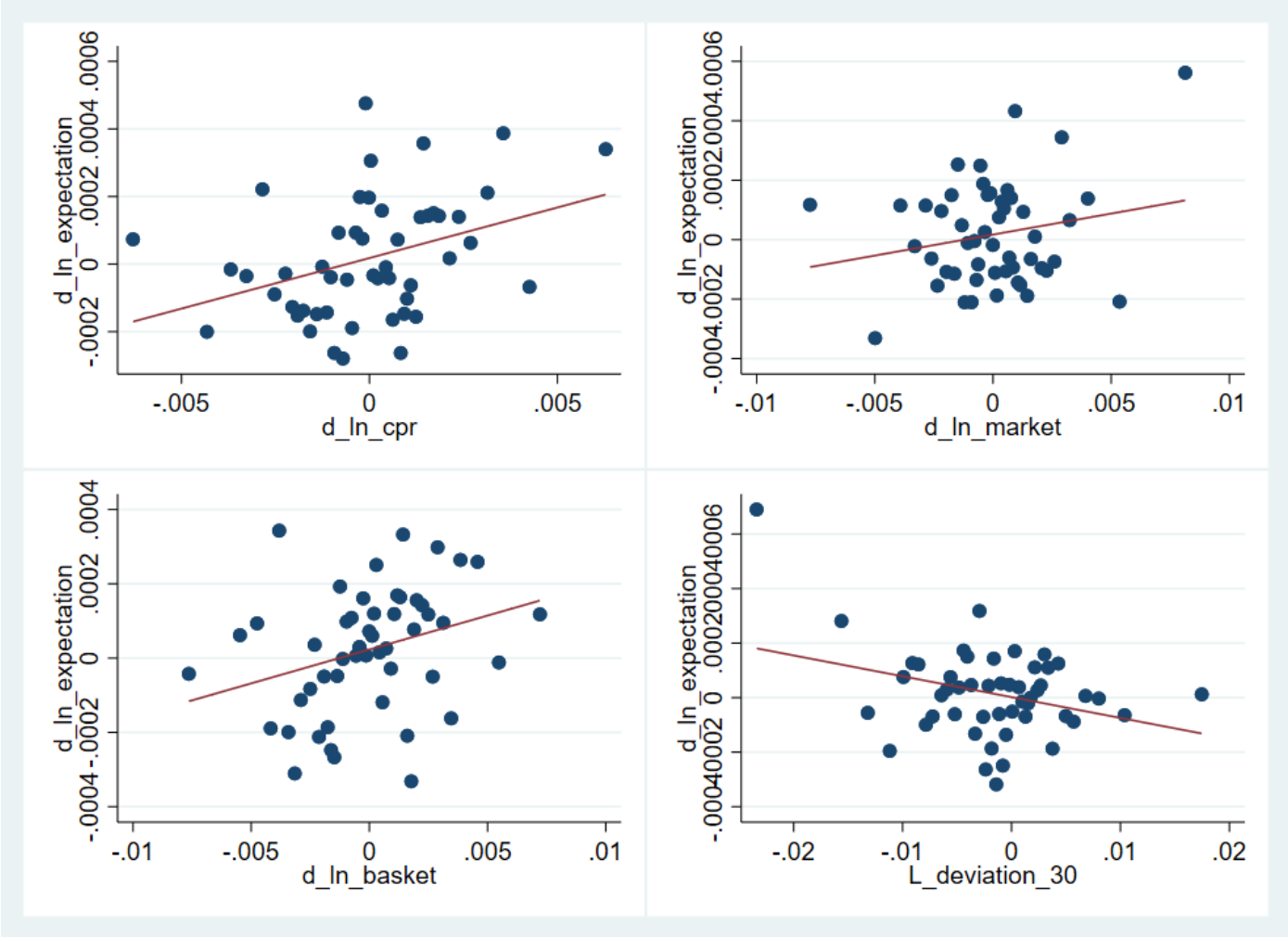
Question: Asymmetric effect of $\widetilde{CPR}$ due to its effect on surveyed FR?

Non-zero revision responds to CPR deviation

	prop_nonzero		
Sample	All	L.deviation<0	L.deviation>0
L.deviation	**-0.224****	**-0.400****	**-0.345***
	(-2.19)	(-2.33)	(-1.80)
L.prop_nonzero	0.156***	0.230***	0.0434
	(3.82)	(4.41)	(0.78)
D.ln_market	0.171	-0.226	0.775**
	(0.74)	(-0.75)	(2.05)
D.ln_basket	0.169	0.454*	-0.169
	(0.76)	(1.65)	(-0.45)
Constant	Yes	Yes	Yes
Observations	898	538	360
R-square	0.029	0.062	0.017

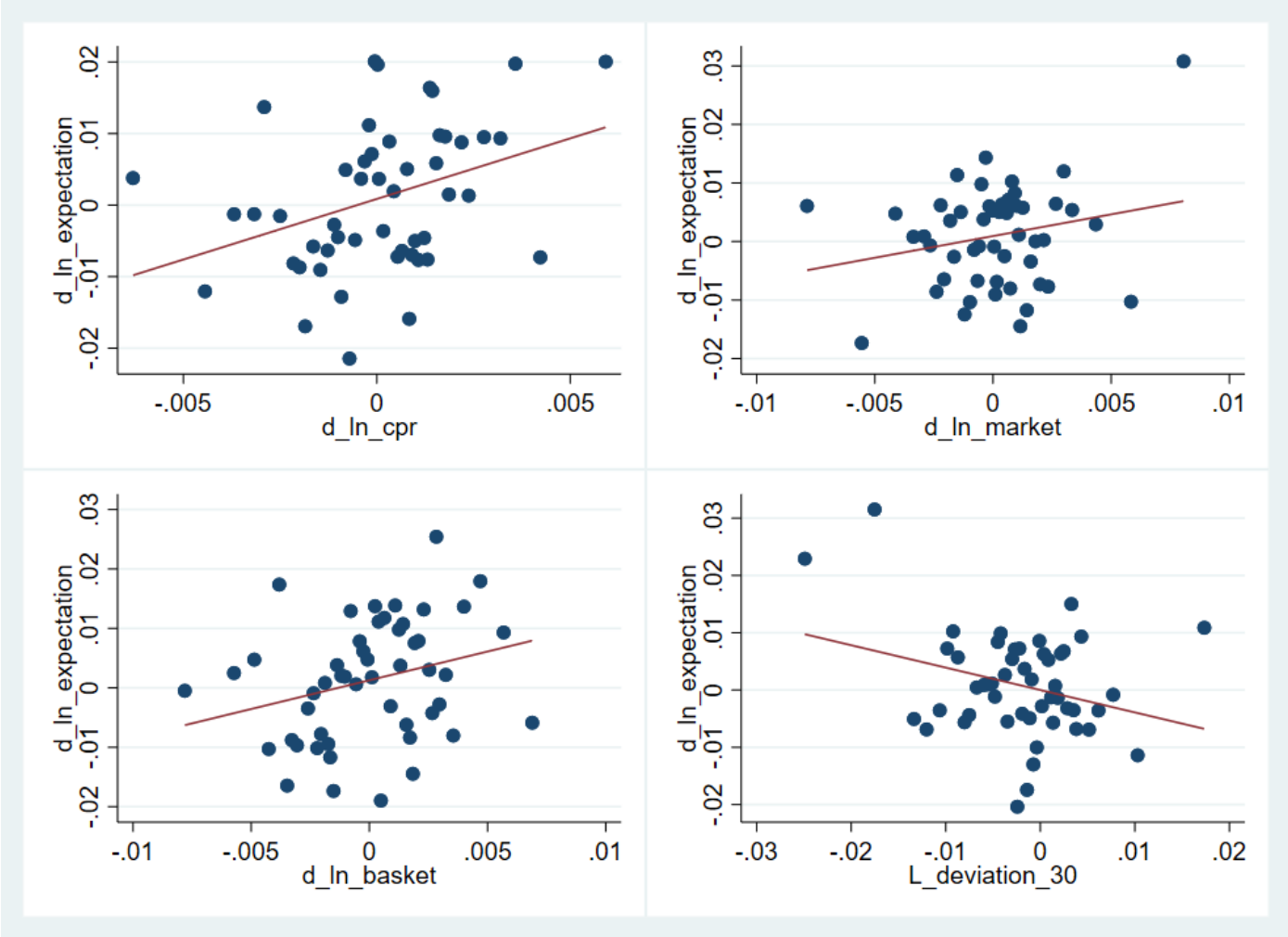
Notes: Here prop_nonzero is the proportion of non-zero revision on each day. The t values based on robust standard errors are included in the parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Forecast revision against pillar changes and CPR deviation



with zero-revisions

Forecast revision against pillar changes and CPR deviation



without zero-revisions

Regression Model

$$F^i_{T|t} - F^i_{T|t-1} = \alpha + \beta \times \widetilde{CPR}_{t-1} + \mathbb{Z}^i_{T,t}\gamma' + \delta_{year} + \mu_i + \mu_T + \varepsilon_{i,T,t}$$

- $F^i_{T|t} - F^i_{T|t-1}$: forecaster i's revision for target T at t;
- $\widetilde{CPR}_{t-1}$: lagged CPR deviation;
- $\mathbb{Z}^i_{T,t}$: forecaster i's prior, changes in pillars, T-t
- δ_{year}: year fixed effect;
- μ_i: forecaster fixed effect;
- μ_T: target fixed effect

Baseline results

	Forecast Revision						
Horizon	pooled	1m-3m	3m-6m	6m-1y	1y-2y	3y-5y	>5y
L.deviation	**-0.014***	**-0.020***	**-0.016****	**-0.013****	**-0.011***	**-0.011****	**-0.018**
	(-3.51)	(-1.97)	(-2.35)	(-2.45)	(-1.94)	(-2.46)	(-1.29)
Prior	-0.006***	-0.037***	-0.021***	-0.013***	-0.007***	-0.005***	-0.002
	(-7.08)	(-11.53)	(-18.41)	(-9.13)	(-7.04)	(-5.62)	(-1.16)
D.ln_market	0.003	-0.012	-0.011	-0.003	0.003	0.000	-0.007
	(0.34)	(-0.94)	(-1.13)	(-0.32)	(0.47)	(0.02)	(-0.31)
D.ln_basket	0.017**	0.016	0.008	0.013	0.014***	0.007	0.013
	(2.49)	(1.35)	(1.55)	(1.65)	(3.92)	(0.65)	(0.60)
length	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	237457	17173	42470	85258	46321	42522	3712
R-square	0.006	0.037	0.022	0.014	0.008	0.005	0.010

Notes: The t values based on clustered standard errors (forecaster/target/date) are included in the parentheses. Fixed effects contain forecaster/target/date. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Baseline results

Forecast revision opposite to CPR deviation!

- Response to common news (Bauer and Swanson 2023)? – No
- Information effect of PBoC (Nakamura and Steinsson 2018)?
  - Intensive margin? – NO
  - Extensive margin: $\widetilde{CPR} < 0$ induces more FR>0

Response to common news?

Forecast revision opposite to CPR deviation could be due to:

CPR deviation and forecast revision both response to news, but...

- Opposite direction only when large and negative CPR deviation
- Positive CPR deviation no effect on forecast revision

Response to common news? probably not

	Forecast Revision			
	Panel A: deviation<0			
L.deviation range	Pooled	<-0.005	<-0.0025	(-0.0025,0)
L.deviation	**-0.027***** (-4.12)	**-0.034***** (-3.11)	**-0.026***** (-2.81)	**0.025** (0.39)
Control	Yes	Yes	Yes	Yes
FE: Forecaster/Target/Year	Yes	Yes	Yes	Yes
Observations	143604	59812	91625	51979
R-square	0.008	0.011	0.009	0.010
	Panel B: deviation>0			
L.deviation range	Pooled	>0.005	>0.0025	(0,0.0025)
L.deviation	**-0.009** (-1.20)	**0.000** (0.02)	**-0.009** (-0.92)	**0.006** (0.11)
Control	Yes	Yes	Yes	Yes
FE: Forecaster/Target/Year	Yes	Yes	Yes	Yes
Observations	93853	26686	52660	41193
R-square	0.008	0.016	0.012	0.011

Information effect?

Forecast revision opposite to CPR deviation could be due to:

CPR deviation reveals information unintended by PBoC

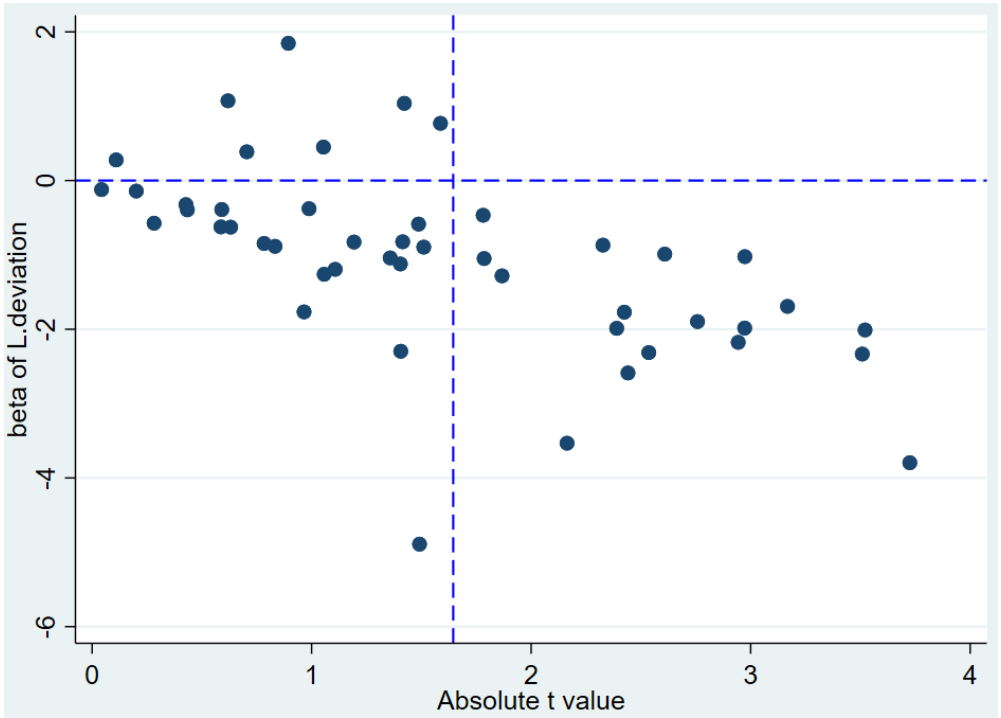
- Individual forecaster regression (excluding zero revisions)

$$F^i_{T|t} - F^i_{T|t-1} = \alpha_i + \beta_i \times \widetilde{CPR}_{t-1} + \mathbb{Z}^i_{T,t}\gamma_i' + \varepsilon_{i,T,t}$$

- Subgroup regression based on FR direction: Against versus Trust
  - Against: FR opposite to $\widetilde{CPR}$
  - Trust: FR align with $\widetilde{CPR}$

Individual forecast revision

Only 18 forecasters respond to $\widetilde{CPR}$ significantly, all negative.



	All sample	Negative response	Positive response
Number of forecasters	47	40	7
Significant 10%	18	18	0

Notes: We have 114 forecasters and 47 of them have # of FR above 24.

Subgroup forecast revision

Conditional on $\widetilde{CPR} < 0$, pooling subgroups yields **stronger** response

	Forecast Revision					
	L.deviation<0			L.deviation>0		
	Pooled	Against	Trust	Pooled	Against	Trust
L.deviation	**-0.893*** **	**-0.177**	**-0.348***	**-0.170**	**-0.638***	**0.469**
	(-4.07)	(-0.92)	(-1.85)	(-0.43)	(-1.80)	(1.68)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2556	1364	1189	1624	841	774
R-square	0.429	0.390	0.457	0.390	0.459	0.416

Notes: The t values based on clustered standard errors (forecaster/target/date) are included in the parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Information effect?

Forecast revision opposite to CPR deviation could be due to:

CPR deviation reveals information unintended by PBoC

- Individual forecaster regression (excluding zero revisions)

$$F^i_{T|t} - F^i_{T|t-1} = \alpha_i + \beta_i \times \widetilde{CPR}_{t-1} + \mathbb{Z}^i_{T,t}\gamma'_i + \varepsilon_{i,T,t}$$

  - Small # of forecasters respond significantly
- Subgroup regression based on FR direction: Against versus Trust
  - Pooling subgroups yields stronger response
- **Take away: Information effect NOT driven by intensive margin.**

Information effect: extensive margin

Trust ratio: % of FRs with same sign as CPR deviation
Against ratio: % of FRs with opposite sign to CPR deviation

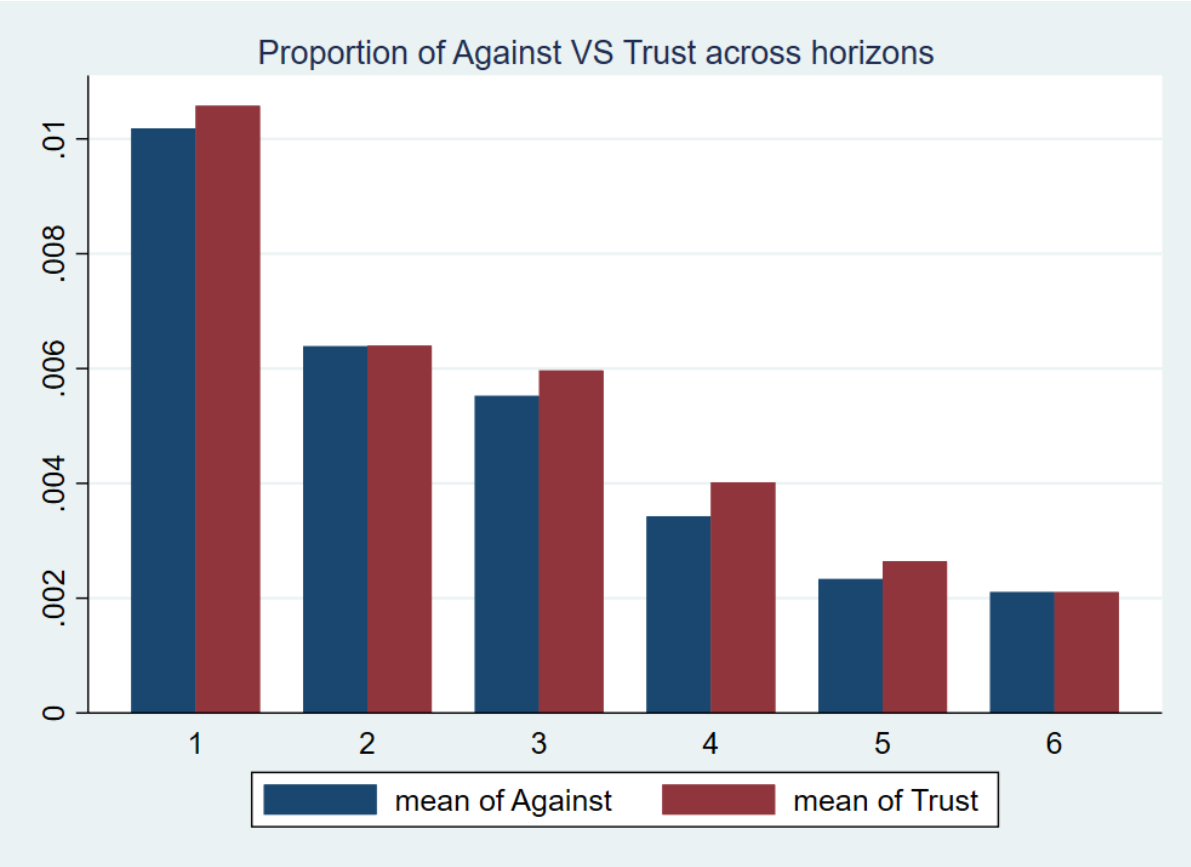


Figure: Proportion of Against VS Trust across horizons

Horizons: 1m-3m, 3m-6m, 6m-1y, 1y-2y, 2y-5y, >5y

Information effect: extensive margin

CPR deviation increases % of Against FRs relative to % of Trust FRs

$$(A-T)_t = \alpha + \beta \times \widetilde{CPR}_{t-1} + \mathbb{Z}_t\gamma' + \delta_{year} + \varepsilon_t$$

- $(A-T)_t$: (Against ratio - Trust ratio) at t;
- $\widetilde{CPR}_{t-1}$: lagged CPR deviation;
- $\mathbb{Z}_t$: $(A-T)_{t-1}$, changes in pillars
- δ_{year}: year fixed effect;

Extensive margin only responds to negative deviation

Against ratio - Trust ratio							
Horizon	pooled	1m-3m	3m-6m	6m-1y	1y-2y	3y-5y	>5y
				Panel A: All			
L.deviation	-0.128	-1.037	-0.069	-0.174	-0.195*	-0.041	0.007
	(-1.09)	(-1.22)	(-0.46)	(-1.29)	(-1.70)	(-0.43)	(0.04)
Observations	898	590	881	898	898	898	835
R-square	0.004	0.006	0.001	0.004	0.004	0.005	0.002
				Panel B: L.deviation<0			
L.deviation	-0.714***	-2.814*	-0.822***	-0.837***	-0.614***	-0.326**	-0.152
	(-3.60)	(-1.95)	(-3.19)	(-3.71)	(-3.25)	(-2.20)	(-0.58)
Observations	538	353	527	538	538	538	507
R-square	0.047	0.048	0.031	0.047	0.025	0.015	0.003
				Panel C: L.deviation>0			
L.deviation	0.139	0.011	-0.044	0.020	0.363	0.264	0.490
	(0.58)	(0.01)	(-0.14)	(0.07)	(1.55)	(1.47)	(1.12)
Observations	360	237	354	360	360	360	328
R-square	0.032	0.090	0.027	0.019	0.022	0.009	0.009

Extensive margin only responds to large negative deviation

	Against ratio - Trust ratio			
	Panel A: L.deviation<0			
L.deviation range	pooled	<-0.005	<-0.0025	(-0.0025,0)
L.deviation	-0.714***	-0.879***	-0.722***	1.134
	(-3.60)	(-2.93)	(-2.88)	(0.54)
Observations	538	226	345	193
R-square	0.047	0.064	0.056	0.014
	Panel B: L.deviation>0			
L.deviation range	pooled	(0,0.0025)	>0.0025	>0.005
L.deviation	0.139	-1.092	0.225	-0.136
	(0.58)	(-0.54)	(0.72)	(-0.43)
Observations	360	158	202	103
R-square	0.032	0.056	0.024	0.003

Notes: The specification is similar to the last table and the results of other variables are omitted for space saving. The *t* values based on robust standard errors are included in the parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Extensive margin responds strongest in bad econ states

		Forecast Revision			Against ratio - Trust ratio		
	State	All	L.dev<0	L.dev>0	All	L.dev<0	L.dev>0
Trade balance	Good	-0.008	-0.002	-0.028*	0.413*	-0.301	0.631*
	Normal	-0.020***	-0.037***	0.003	-0.217	-0.700***	0.067
	Bad	**-0.014****	**-0.033*****	**0.008**	**-0.499*****	**-0.860*****	**-0.630***
Capital flow	Good	0.000	-0.005	-0.004	0.366*	0.097	0.090
	Normal	-0.008	-0.009	-0.022**	0.333	-0.071	0.447
	Bad	**-0.028*****	**-0.041*****	**-0.015**	**-0.771*****	**-1.353*****	**0.201**
Inflation diff	Good	-0.009	-0.022*	-0.006	-0.055	-0.548*	-0.088
	Normal	-0.007	-0.017*	-0.016	-0.217	-0.822***	0.107
	Bad	**-0.022*****	**-0.036*****	**-0.011**	**-0.129**	**-0.726*****	**0.460**
EPU diff	Good	-0.010	-0.021**	-0.007	0.193	-0.281	-0.113
	Normal	-0.007	-0.015**	-0.012	-0.036	-0.558***	0.173
	Bad	**-0.024*****	**-0.040*****	**-0.007**	**-0.404***	**-1.101*****	**0.179**
Dollar index	Good	-0.004	0.032***	-0.048***	0.980***	0.604**	0.816***
	Normal	-0.004	-0.016	0.004	-0.215	-0.530**	-0.446*
	Bad	**-0.030*****	**-0.057*****	**0.035****	**-0.772*****	**-1.152*****	**-0.260**
VIX	Good	-0.003	-0.013	-0.004	0.284	-0.229	0.211
	Normal	-0.016**	-0.029***	-0.004	-0.561***	-1.184***	-0.206
	Bad	**-0.021****	**-0.037*****	**-0.019**	**-0.013**	**-0.661*****	**0.439**

▸ fundamental correlation

Conclusions

PBoC's CPR deviations from the policy rule $\widetilde{CPR}$

- are policy interventions with announcement effect
  - $\widetilde{CPR} > 0$: predicts future CNY/USD change
  - $\widetilde{CPR} < 0$: no predictive power
- has negative information effect on forecast revisions
  - $\widetilde{CPR} > 0$: little effect on FR
  - $\widetilde{CPR} < 0$: induces more FR$>$0 $\rightarrow$ weakens announcement effect

We identify unintended information effect of policy intervention: asymmetric, state-dependent, extensive-margin driven.

CPR and spot rates

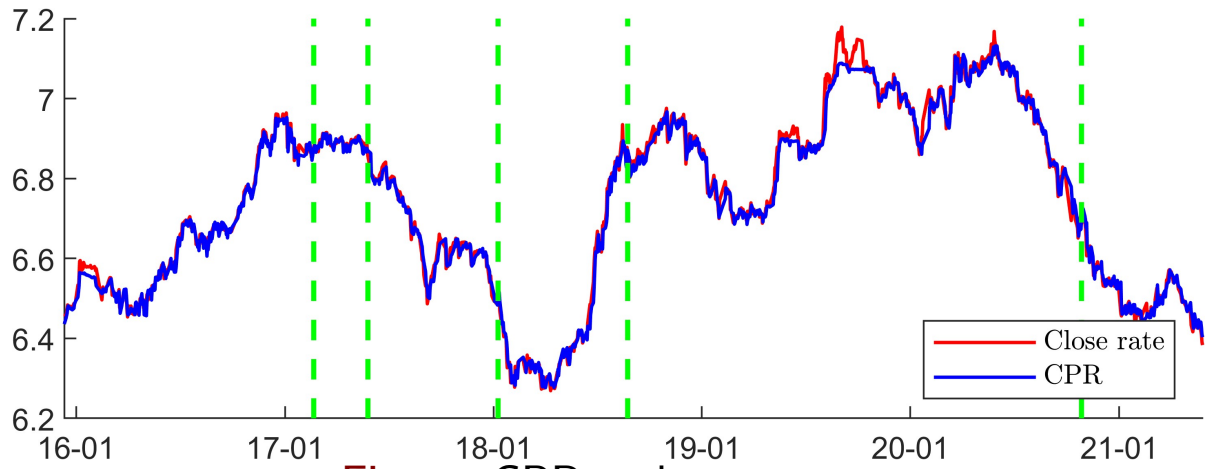


Figure: CPR and spot rates

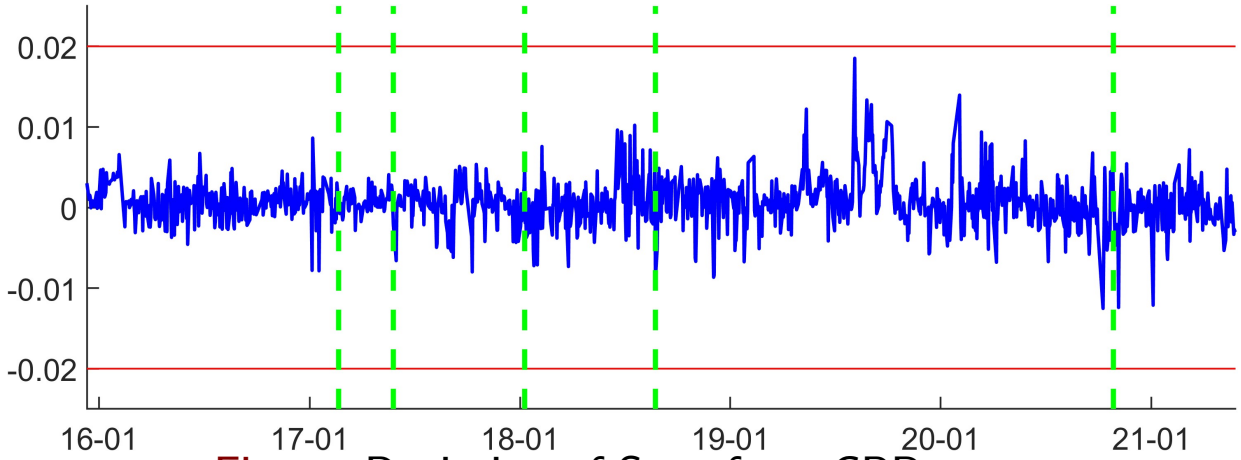
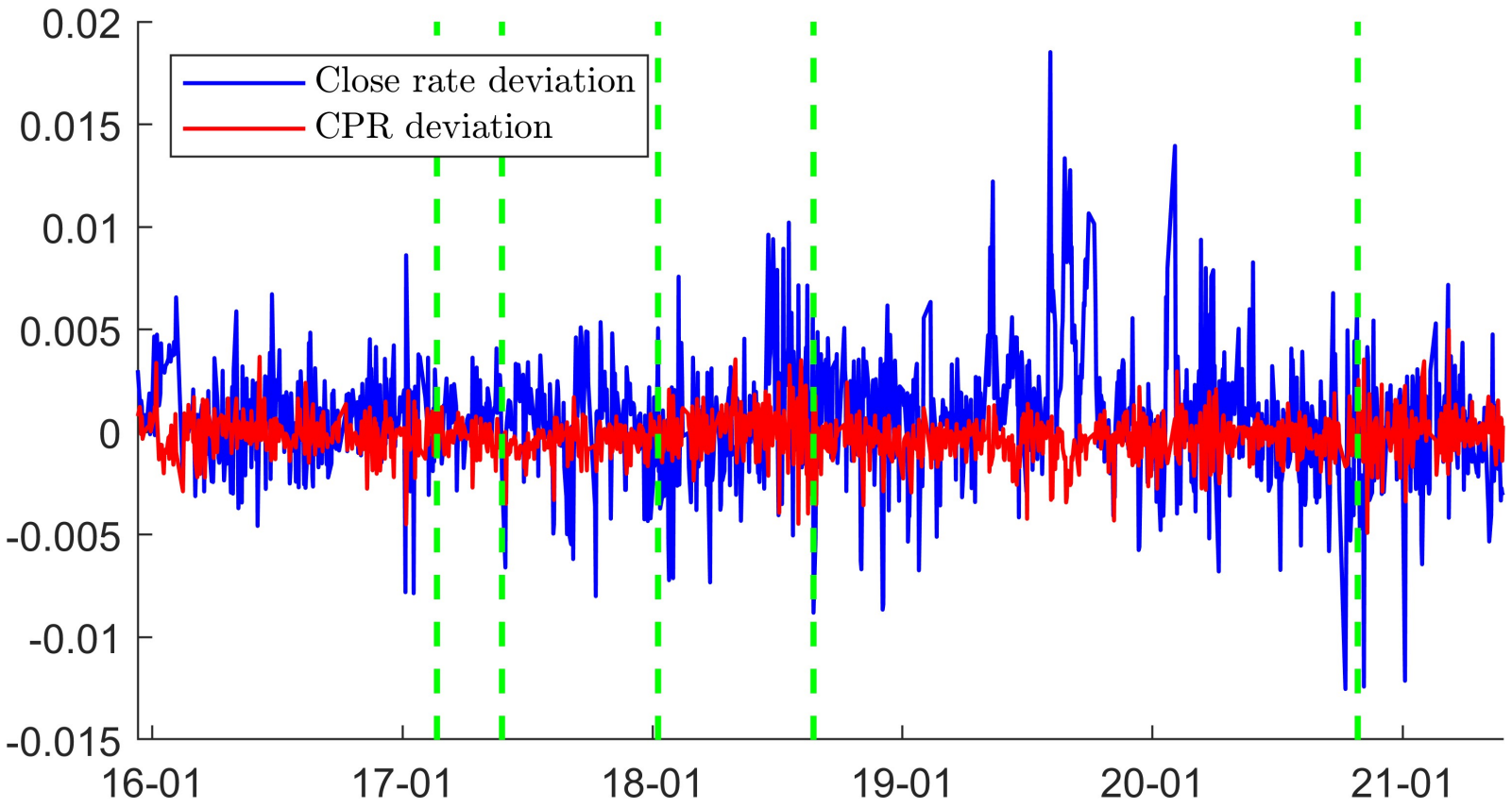


Figure: Deviation of Spot from CPR rates

Back

CPR deviation and Close rate deviation



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CPR deviation not predict future CPR deviation

ave_future_deviation_Xd	1d	2d	3d	4d	5d	10d
			Panel A: all sample			
deviation	**-0.037**	**-0.006**	**0.011**	**0.015**	**0.025**	**0.020***
	(-0.99)	(-0.25)	(0.49)	(0.78)	(1.51)	(1.80)
Observations	1131	1125	1119	1113	1107	1077
R-square	0.038	0.070	0.102	0.130	0.154	0.235
			Panel B: deviation<0			
deviation	**0.002**	**0.010**	**0.002**	**0.019**	**0.017**	**-0.005**
	(0.03)	(0.21)	(0.06)	(0.56)	(0.56)	(-0.24)
Observations	679	676	672	668	666	649
R-square	0.047	0.075	0.108	0.132	0.151	0.218
			Panel C: deviation>0			
deviation	**-0.186***	**-0.117****	**-0.055**	**-0.068**	**-0.041**	**-0.015**
	(-1.77)	(-1.98)	(-1.03)	(-1.64)	(-1.07)	(-0.63)
Observations	452	449	447	445	441	428
R-square	0.041	0.074	0.091	0.125	0.148	0.242

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CPR deviation not predict future CPR deviation

ave_future_deviation_Xd	20d	30d	40d	50d	60d	120d
			Panel A: all sample			
deviation	**0.021****	**0.015***	**0.010**	**0.009**	**0.011***	**0.006**
	(2.35)	(1.93)	(1.53)	(1.50)	(1.83)	(1.31)
Observations	1017	957	953	953	953	875
R-square	0.349	0.393	0.405	0.393	0.392	0.254
			Panel B: deviation<0			
deviation	**0.011**	**-0.009**	**-0.015**	**-0.016**	**-0.012**	**-0.003**
	(0.62)	(-0.63)	(-1.21)	(-1.36)	(-1.07)	(-0.30)
Observations	613	577	571	574	578	538
R-square	0.302	0.344	0.354	0.335	0.327	0.205
			Panel C: deviation>0			
deviation	**0.015**	**0.014**	**0.008**	**0.024****	**0.027****	**-0.001**
	(0.87)	(0.96)	(0.59)	(2.06)	(2.31)	(-0.05)
Observations	404	380	382	379	375	337
R-square	0.397	0.442	0.459	0.456	0.468	0.315

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Forecast revision explains exchange rate change

$$S^{CL}_{t+j} - S^{CL}_{t-1} = \alpha + \beta \sum_{T,i} \frac{F^{i}_{T|t+j} - F^{i}_{T|t-1}}{N_T N_i} + \mathbb{Z}_t \gamma' + \varepsilon_t; \tag{4}$$

dln_close_Xd	1d	2d	3d	4d	5d	10d
Forecast revision	**0.704***** (6.31)	**0.697***** (8.55)	**0.676***** (10.51)	**0.662***** (12.47)	**0.667***** (13.69)	**0.770***** (18.75)
length	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1148	1147	1146	1145	1144	1139
R-square	0.135	0.149	0.161	0.168	0.183	0.299

dln_close_Xd	20d	30d	40d	50d	60d	120d
Forecast revision	**0.878***** (28.46)	**0.912***** (37.96)	**0.900***** (43.97)	**0.877***** (49.01)	**0.851***** (53.97)	**0.782***** (69.44)
length	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1129	1119	1109	1099	1089	1029
R-square	0.504	0.621	0.694	0.741	0.778	0.855

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Fundamental correlation

	Trade balance	Capital flow	Inflation diff	EPU diff	Dollar index	VIX
Trade balance	1.00					
Capital flow	-0.25	1.00				
Inflation diff	0.03	-0.31	1.00			
EPU diff	-0.37	0.11	-0.26	1.00		
Dollar index	0.17	-0.20	0.46	-0.56	1.00	
VIX	0.52	-0.05	0.23	-0.74	0.48	1.00

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R^2 comparison: deviation

dln_close_Xd	1d	2d	3d	4d	5d	10d	20d	30d	40d	50d	60d	120d
						Panel A: all sample						
with deviation	0.036	0.036	0.029	0.023	0.026	0.038	0.086	0.154	0.197	0.221	0.23	0.329
without deviation	0.028	0.028	0.02	0.018	0.022	0.034	0.085	0.149	0.194	0.218	0.228	0.329
change(%)	28.57	28.57	45.00	27.78	18.18	11.76	1.18	3.36	1.55	1.38	0.88	0.00
						Panel B: deviation<0						
with deviation	0.03	0.025	0.016	0.014	0.014	0.025	0.054	0.123	0.143	0.139	0.137	0.212
without deviation	0.027	0.025	0.016	0.014	0.014	0.024	0.052	0.123	0.142	0.139	0.137	0.211
change(%)	11.11	0.00	0.00	0.00	0.00	4.17	3.85	0.00	0.70	0.00	0.00	0.47
						Panel C: deviation>0						
with deviation	0.028	0.034	0.022	0.023	0.035	0.045	0.129	0.164	0.249	0.305	0.349	0.488
without deviation	0.025	0.022	0.014	0.017	0.026	0.038	0.126	0.161	0.246	0.304	0.339	0.483
change(%)	12.00	54.55	57.14	35.29	34.62	18.42	2.38	1.86	1.22	0.33	2.95	1.04

R^2 comparison: forecast revision

dln_close_Xd	1d	2d	3d	4d	5d	10d
with revision	0.135	0.079	0.058	0.049	0.052	0.058
without revision	0.002	0.005	0.008	0.011	0.013	0.026
change(%)	6650.00	1480.00	625.00	345.45	300.00	123.08

dln_close_Xd	20d	30d	40d	50d	60d	120d
with revision	0.091	0.129	0.177	0.233	0.289	0.465
without revision	0.057	0.101	0.155	0.215	0.275	0.463
change(%)	59.65	27.72	14.19	8.37	5.09	0.43

Control interest rate: revision on deviation

	Forecast Revision					
	All sample		deviation<0		deviation>0	
L.deviation	-0.014***	-0.014***	-0.027***	-0.027***	-0.009	-0.009
	(-3.51)	(-3.56)	(-4.12)	(-4.14)	(-1.20)	(-1.21)
d.diffint	No	Yes	No	Yes	No	Yes
Control	Yes	Yes	Yes	Yes	Yes	Yes
Observations	237457	237194	143604	143341	93853	93853
R-square	0.006	0.006	0.008	0.008	0.008	0.008

Control interest rate: close rate on deviation

dln_close_Xd	1d	2d	3d	4d	5d	10d
			Panel A: all sample			
deviation	**0.028****	**0.048*****	**0.062*****	**0.050****	**0.053****	**0.079****
	(2.04)	(2.68)	(2.99)	(2.06)	(2.01)	(2.15)
Observations	1099	1093	1087	1081	1076	1046
R-square	0.037	0.039	0.032	0.022	0.024	0.036
			Panel B: deviation<0			
deviation	**0.010**	**-0.012**	**0.004**	**-0.017**	**-0.015**	**-0.048**
	(0.43)	(-0.40)	(0.12)	(-0.39)	(-0.31)	(-0.75)
Observations	660	657	653	649	648	631
R-square	0.035	0.029	0.017	0.013	0.013	0.021
			Panel C: deviation>0			
deviation	**0.039**	**0.100****	**0.100***	**0.097***	**0.121****	**0.170***
	(1.15)	(2.38)	(1.91)	(1.74)	(1.99)	(1.95)
Observations	439	436	434	432	428	415
R-square	0.034	0.040	0.029	0.026	0.036	0.050

Control interest rate: close rate on deviation

dln_close_Xd	20d	30d	40d	50d	60d	120d
			Panel A: all sample			
deviation	**0.062**	**0.206****	**0.134**	**0.159***	**0.133**	**0.075**
	(1.10)	(2.57)	(1.49)	(1.65)	(1.31)	(0.51)
Observations	990	931	922	920	919	846
R-square	0.083	0.154	0.195	0.223	0.229	0.322
			Panel B: deviation<0			
deviation	**-0.102**	**0.030**	**0.040**	**0.055**	**-0.064**	**0.180**
	(-1.09)	(0.22)	(0.29)	(0.37)	(-0.42)	(0.75)
Observations	597	561	552	553	557	520
R-square	0.051	0.120	0.145	0.150	0.147	0.215
			Panel C: deviation>0			
deviation	**0.179**	**0.271**	**0.234**	**0.165**	**0.676****	**0.624**
	(1.23)	(1.19)	(0.82)	(0.56)	(2.46)	(1.55)
Observations	393	370	370	367	362	326
R-square	0.127	0.166	0.249	0.301	0.337	0.468